

A Formalization of Kakutani’s Dichotomy for Infinite Product Measures

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Abstract

Kakutani’s dichotomy (1948) states that an infinite product of probability measures, built from factors pairwise locally absolutely continuous, is either absolutely continuous or mutually singular with respect to a second such product, according to whether the infinite product of the Hellinger affinities of the factor pairs is positive or zero. We present a formalization of both directions of the theorem in Lean 4 over Mathlib, stated for families indexed by an arbitrary type, and with no absolute-continuity hypothesis at all in the singular direction; both are strengthenings of the classical statement, and to the authors’ knowledge this is the first formalization of the theorem in any proof assistant. The development defines the Hellinger affinity as a total $\mathbb{R}_{\geq 0}^\infty$ -valued function of a pair of measures, proves the singular direction by a cylinder squeeze and the Borel–Cantelli lemma, and proves the absolutely continuous direction by a quantitative comparison of the product measure with a finite-window density measure over the measure-dense algebra of cylinders. Martingale convergence, the Kolmogorov zero–one law, and L^p -completeness arguments are not used. Two applications are reported. A family of product Poisson measures on an infinite-dimensional torus indexed by the primes is shown to be mutually singular with product Haar measure exactly for parameters $\sigma \leq 1/2$ and equivalent to it exactly for $\sigma > 1/2$; and for every nonzero shift $|u| \leq \pi/(3 \log 4)$ the log-scale translate of the same family is shown to be mutually singular with the family itself throughout $0 < \sigma \leq 1/2$, endpoint included, from Chebyshev prime bounds alone. The development carries a zero-sorry continuous-integration policy, and the main theorems depend only on the axioms `propext`, `Classical.choice`, and `Quot.sound`. Upstreaming to Mathlib is planned as a sequence of six pull requests.

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Supplementary Material *Software (Source Code)*: <https://github.com/idolum-ai/riemann-venue>

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1 Introduction

Let $(\Omega_i)_{i \in I}$ be measurable spaces carrying two families of probability measures μ_i and ν_i with $\mu_i \ll \nu_i$ for every i , and let $\mu = \bigotimes_i \mu_i$ and $\nu = \bigotimes_i \nu_i$ be the product measures on $\prod_i \Omega_i$. Kakutani’s theorem [11] says that the relationship between μ and ν is decided by a single numerical invariant, the infinite product of the Hellinger affinities $H(\mu_i, \nu_i) = \int \sqrt{\frac{d\mu_i}{d\nu_i}} d\nu_i$: if $\prod_i H(\mu_i, \nu_i) > 0$ then $\mu \ll \nu$, and if the product is 0 then μ and ν are mutually singular.



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When each pair is mutually absolutely continuous, the conclusion sharpens to a dichotomy between equivalence and mutual singularity, with no intermediate regime. The formalization presented here proves the singular direction without the hypothesis $\mu_i \ll \nu_i$; the affinity is defined so as to be total, and singular factor parts only help singularity (Subsection 3.4).

The theorem is a standard instrument of measure-theoretic probability. It is the ancestor of the Feldman–Hájek dichotomy for Gaussian measures [5, 6], it governs when two stochastic models with independent coordinates can be distinguished with certainty from a single sample, and its criterion is effective: since $0 \leq H(\mu_i, \nu_i) \leq 1$, the product is positive precisely when the deficits $1 - H(\mu_i, \nu_i)$ are summable, so verifying either branch of the dichotomy reduces to a convergence estimate.

To our knowledge the theorem had no formalization in any proof assistant before this work; Section 6 and Section 7 record the basis for this claim and the protocol for re-verifying it prior to publication. One structural reason is that the statement needs infinite products of probability measures over arbitrary index sets as first-class objects, and Lean's Mathlib [13, 3] acquired these only recently (`Measure.infinitePi`, built through the Ionescu–Tulcea theorem [10]). A second reason is that the textbook proofs [14] run through martingale convergence and L^2 limit arguments, which are formalizable but drag in machinery whose bookkeeping dominates the mathematics. The routes chosen here avoid that machinery entirely.

The contributions:

- Machine-checked statements and proofs of both directions of Kakutani's dichotomy in Lean 4 over Mathlib, for factor families indexed by an *arbitrary* type, and with no absolute-continuity hypothesis in the singular direction. Both strengthenings of the classical countable statement fell out of the proof routes and the definitions rather than being goals.
- A reusable definition of the Hellinger affinity as a total, $\mathbb{R}_{\geq 0}^\infty$ -valued, symmetric function of a pair of measures, with its basic API, including $H(\mu, \nu) = 0 \leftrightarrow \mu \perp \nu$ (Subsection 3.2).
- Supporting lemmas absent from Mathlib at the pinned revision: finite tensorization of densities (`Measure.pi_withDensity`), a product Fubini theorem for the lower Lebesgue integral, positivity and summability bridges for infinite products in $\mathbb{R}_{\geq 0}^\infty$, and window-factorization lemmas over `Measure.infinitePi` (Section 3).
- Proof routes suited to formalization: the singular direction by a cylinder squeeze and Borel–Cantelli, the absolutely continuous direction by a quantitative cylinder comparison. Neither uses martingales, the zero–one law, or L^p completeness (Section 4).
- Two applications: the family of product Poisson measures over the primes at ratios $p^{-\sigma}$ changes type against product Haar measure exactly at $\sigma = 1/2$; and its log-scale translate is mutually singular with the family itself for all $0 < \sigma \leq 1/2$ and explicit nonzero shifts, endpoint included, with no prime number theorem input (Section 5).
- A public artifact with zero-sorry CI, axiom audits, and a staged plan for upstreaming the general-purpose layer to Mathlib (Section 7).

The formalization was built as an instrument for a research program on the spectral geometry of the greatest-common-divisor kernel [12]; Section 5 states the two theorems of that program that this paper reports.

2 Mathematical background

2.1 The Hellinger affinity

For probability measures μ, ν on a measurable space Ω and any σ -finite measure λ dominating both, the *Hellinger affinity* (also the Bhattacharyya coefficient [7, 2]) is

$$H(\mu, \nu) = \int_{\Omega} \sqrt{\frac{d\mu}{d\lambda} \cdot \frac{d\nu}{d\lambda}} d\lambda,$$

independent of the choice of λ . It is symmetric, and by Cauchy–Schwarz $0 \leq H(\mu, \nu) \leq 1$, with $H(\mu, \mu) = 1$. The two endpoints carry the geometry: $H(\mu, \nu) = 1$ iff $\mu = \nu$, and $H(\mu, \nu) = 0$ iff $\mu \perp \nu$. The affinity tensorizes over finite products, $H(\mu_1 \otimes \mu_2, \nu_1 \otimes \nu_2) = H(\mu_1, \nu_1) \cdot H(\mu_2, \nu_2)$, which is what makes it the right invariant for product measures.

2.2 The dichotomy

► **Theorem 1** (Kakutani [11]). *For each i in a countable index set let μ_i, ν_i be probability measures on Ω_i with $\mu_i \ll \nu_i$, and set $\mu = \bigotimes_i \mu_i$, $\nu = \bigotimes_i \nu_i$.*

1. *If $\prod_i H(\mu_i, \nu_i) > 0$, then $\mu \ll \nu$.*

2. *If $\prod_i H(\mu_i, \nu_i) = 0$, then $\mu \perp \nu$.*

If moreover $\nu_i \ll \mu_i$ for every i , the first case gives equivalence of μ and ν .

Since every factor lies in $[0, 1]$, the partial products are antitone and the infinite product converges to a limit in $[0, 1]$; positivity of the limit is equivalent to summability of the deficits:

$$\prod_i H(\mu_i, \nu_i) > 0 \iff \sum_i (1 - H(\mu_i, \nu_i)) < \infty \text{ and } H(\mu_i, \nu_i) > 0 \text{ for all } i.$$

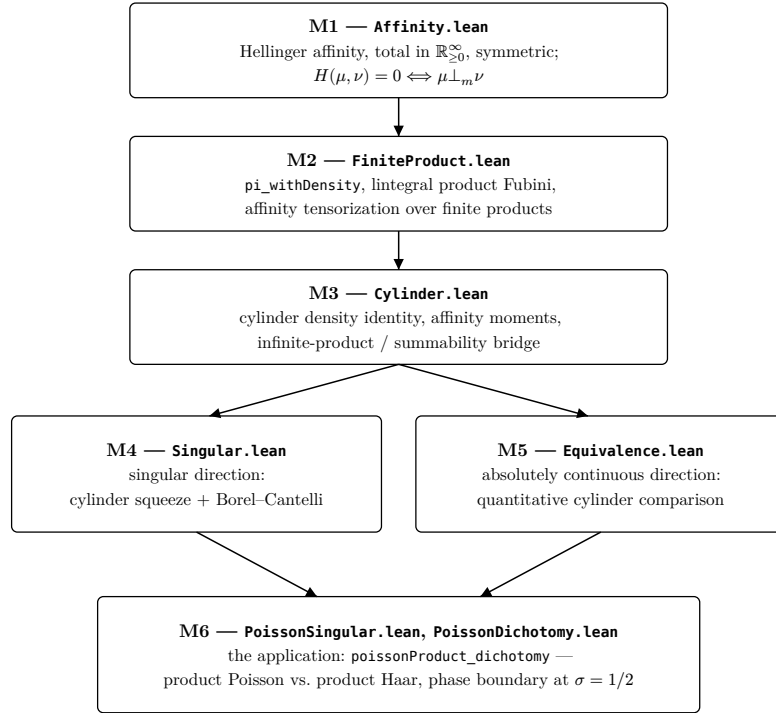
The classical proof [11, 14] considers the likelihood ratios $Z_n = \prod_{i \leq n} \frac{d\mu_i}{d\nu_i}$, a nonnegative ν -martingale. In the convergent case the square roots $\sqrt{Z_n}$ form a Cauchy sequence in $L^2(\nu)$, the limit identifies $\frac{d\mu}{d\nu}$, and absolute continuity follows; in the divergent case $Z_n \rightarrow 0$ almost everywhere under ν while μ concentrates where Z_n survives, and a zero–one argument upgrades this to mutual singularity. The formalized proofs reach the same conclusions along different routes (Section 4), and the singular one without the hypothesis $\mu_i \ll \nu_i$.

3 The formalization

The development comprises seven general-purpose Lean files plus five application files under `RiemannVenue/Kakutani/` in the artifact [12], built against a pinned Mathlib revision (Section 7). Everything in this section and the next lives in the `MeasureTheory` namespace, most of it under `Measure`; the application (Section 5) lives in the repository’s own namespace. Figure 1 shows the milestone ladder; each box was designed, then implemented as one complete file with no placeholders.

3.1 Infinite product measures in Mathlib

Mathlib’s `Measure.infinitePi` takes an arbitrary index type ι and a family $\mu : (i : \iota) \rightarrow \text{Measure}(X\ i)$ of probability measures, and produces the product probability measure on $\prod_i X\ i$. The N-indexed core is the projective limit supplied by the Ionescu–Tulcea theorem [10]; the general case is a Carathéodory extension of the induced content on measurable cylinders. The API this development consumes is the cylinder evaluation (`infinitePi_`



■ **Figure 1** The milestone ladder. M1–M3 are infrastructure, M4 and M5 are the two directions of the dichotomy (independent of one another), M6 is the application. Arrows are import dependencies.

125 cylinder), the finite-marginal identification (`infinitePi_map_restrict`), and the restric-
 126 tion rule for lower Lebesgue integrals of finite-window functions (`integral_restrict_`
 127 `infinitePi`). The recency of this layer is the main reason a 1948 theorem was still open to
 128 formalization.

129 3.2 The Hellinger affinity as formalized

```

130
131 /-- The Hellinger affinity (Bhattacharyya coefficient) of two measures:
132 '∫ √(dμ/dσ · dν/dσ) dσ' computed against the canonical dominating measure
133 'σ = μ + ν'. Total and symmetric; no absolute-continuity hypothesis. -/
134 noncomputable def hellingerAffinity (μ ν : Measure Ω) : ℝ≥0∞ :=
135   ∫- x, (μ.rnDeriv (μ + ν) x * ν.rnDeriv (μ + ν) x) ^ (1 / 2 : ℝ) ∂(μ + ν)
136

```

137 Three design decisions are packed into this definition. First, the dominating measure
 138 is fixed to $\mu + \nu$, so the affinity is total: it needs no absolute-continuity or integrability
 139 hypotheses, and an invariance lemma (`hellingerAffinity_eq_integral_rnDeriv_mul_`
 140 `rnDeriv`) recovers the value against any dominating σ -finite measure. Second, the value
 141 type is $\mathbb{R}_{\geq 0}^\infty$ and the integral is the lower Lebesgue integral, so no side conditions about in-
 142 tegrability arise anywhere in the development; Mathlib's Kullback–Leibler divergence `kldiv`
 143 made the same choice. Third, totality is what makes the singular direction of the dichotomy
 144 free of case splits: the characterization

```

145
146 theorem hellingerAffinity_eq_zero_iff [IsFiniteMeasure μ] [IsFiniteMeasure ν] :
147   hellingerAffinity μ ν = 0 ↔ μ ⊥m ν
148

```

holds with no hypotheses beyond finiteness, so degenerate factor pairs need no separate treatment. The remaining API proves symmetry, $H(\mu, \mu) = 1$, $H(\mu, \nu) \leq 1$ (Hölder with exponents 2, 2), positivity ($H(\mu, \nu) > 0 \leftrightarrow \neg \mu \perp \nu$), monotonicity in each argument, the `withDensity` computation rule, tensorization over finite products (`hellingerAffinity_pi`), and the identity $H(\mu, \nu) = \int (\frac{d\mu}{d\nu})^{1/2} d\nu$ for arbitrary finite measures: by Lebesgue decomposition the singular part of μ has its $(\mu + \nu)$ -density supported where the density of ν vanishes, so it contributes nothing to the defining integral (`hellingerAffinity_eq_lintegral_rnDeriv`). The totality of this identity is what later makes the moment identity of the singular route hypothesis-free.

3.3 The hypothesis carrier: infima of partial products

The dichotomy's hypothesis is the value of an infinite product. Mathlib's `tprod` is unsuitable as a carrier for it: like `tsum`, it returns the junk value 1 for non-multipliable families, and “the infinite product is 0” is exactly the case where multipliability fails. Since every factor is ≤ 1 , the finite partial products form an antitone net, and its infimum is the honest value of the infinite product. The formalized statements therefore quantify over finite subsets:

$$\prod_{s:\text{Finset } \iota} \prod_{i \in s} H(\mu_i, \nu_i) = 0 \quad \text{versus} \quad 0 < \prod_s \prod_{i \in s} H(\mu_i, \nu_i).$$

A bridge lemma converts between this form and the summability criterion:

```

theorem iInf_finsetProd_pos_iff_summable_one_sub {h : κ → ℝ≥0∞}
  (h1 : ∀ i, h i ≤ 1) :
  (0 < ∏ s : Finset κ, ∏ i ∈ s, h i)
  ↔ (∀ i, h i ≠ 0) ∧ Summable fun i => 1 - (h i).toReal

```

Its proof rests on two elementary inequalities, formalized over the reals: the Weierstrass bound $1 - \prod_{i \in s} h_i \leq \sum_{i \in s} (1 - h_i)$ and the exponential bound $\prod_{i \in s} h_i \leq \exp(-\sum_{i \in s} (1 - h_i))$.

3.4 The main statements

The following are quoted verbatim from the artifact. The ambient context is

```

variable {ι : Type*} {X : ι → Type*} {mX : ∀ i, MeasurableSpace (X i)}
(μ ν : (i : ι) → Measure (X i))
[∀ i, IsProbabilityMeasure (μ i)] [∀ i, IsProbabilityMeasure (ν i)]

```

with no constraint on ι . The singular direction (`Singular.lean`), with no hypothesis on the factor pairs at all:

```

theorem infinitePi_mutuallySingular
  (h : ∏ s : Finset ι, ∏ i ∈ s, hellingerAffinity (μ i) (ν i) = 0) :
  Measure.infinitePi μ ⊥m Measure.infinitePi ν

```

The absolutely continuous direction, in its summability and infimum forms, and the two-sided corollary under mutual local absolute continuity (`Equivalence.lean`):

```

theorem infinitePi_absolutelyContinuous_of_summable (hac : ∀ i, μ i ≪ ν i)
  (hsum : Summable fun i => 1 - (hellingerAffinity (μ i) (ν i)).toReal) :
  Measure.infinitePi μ ≪ Measure.infinitePi ν

theorem infinitePi_absolutelyContinuous (hac : ∀ i, μ i ≪ ν i)
  (h : 0 < ∏ s : Finset ι, ∏ i ∈ s, hellingerAffinity (μ i) (ν i)) :

```

```

197   Measure.infinitePi  $\mu \ll$  Measure.infinitePi  $\nu$ 
198
199   theorem infinitePi_absolutelyContinuous_and_symm
200     (hac :  $\forall i, \mu i \ll \nu i$ ) (hac' :  $\forall i, \nu i \ll \mu i$ )
201     (h :  $0 < \prod s : \text{Finset } \iota, \prod i \in s, \text{hellingerAffinity } (\mu i) (\nu i)$ ) :
202     Measure.infinitePi  $\mu \ll$  Measure.infinitePi  $\nu \wedge$ 
203     Measure.infinitePi  $\nu \ll$  Measure.infinitePi  $\mu$ 

```

The packaged forms (Dichotomy.lean): the two-sided dichotomy, and the criterion as a biconditional,

```

207   theorem infinitePi_absolutelyContinuous_or_mutuallySingular
208     (hac :  $\forall i, \mu i \ll \nu i$ ) (hac' :  $\forall i, \nu i \ll \mu i$ ) :
209     (Measure.infinitePi  $\mu \ll$  Measure.infinitePi  $\nu \wedge$ 
210      Measure.infinitePi  $\nu \ll$  Measure.infinitePi  $\mu$ )  $\vee$ 
211     Measure.infinitePi  $\mu \perp_m$  Measure.infinitePi  $\nu$ 
212
213   theorem infinitePi_absolutelyContinuous_iff (hac :  $\forall i, \mu i \ll \nu i$ ) :
214     Measure.infinitePi  $\mu \ll$  Measure.infinitePi  $\nu \leftrightarrow$ 
215      $0 < \prod s : \text{Finset } \iota, \prod i \in s, \text{hellingerAffinity } (\mu i) (\nu i)$ 

```

Both are order splits on the infimum, exhaustive in $\mathbb{R}_{\geq 0}^{\infty}$ with no case analysis debt. The biconditional needs only the one-sided hypothesis, which is sharper than the design asked for: its forward implication is exclusivity through the hypothesis-free singular direction, so weakening the singular statement once weakened every downstream packaging for free.

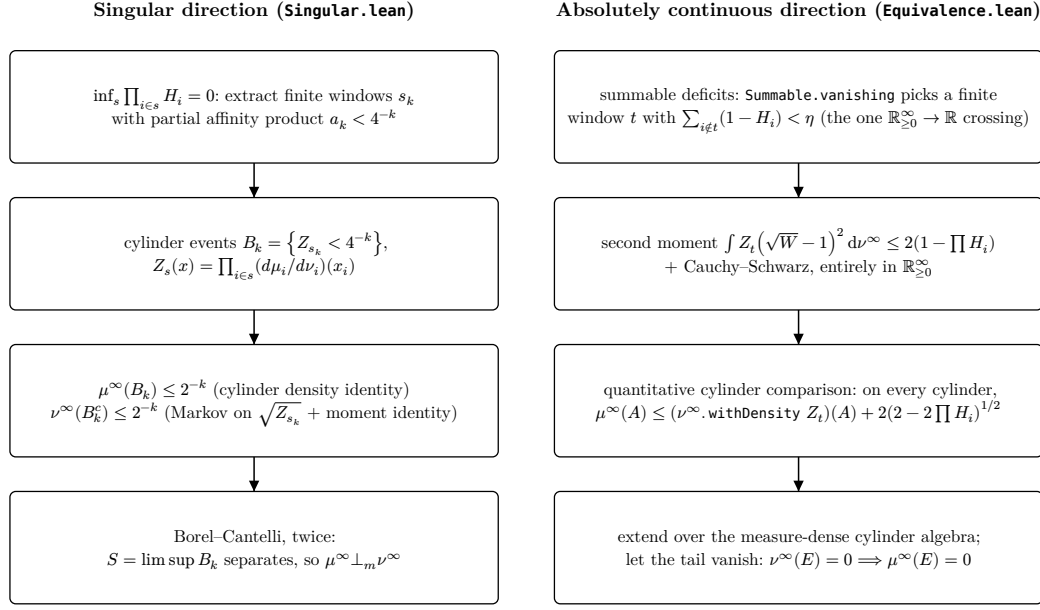
Two remarks on the hypotheses, in the interest of precision. The index type is arbitrary in both directions; Section 4 explains why countability is never needed. And the asymmetry between the directions is exact. The absolutely continuous direction keeps **hac**, as it must: a single factor with a nonzero singular part destroys $\mu \ll \nu$ while the affinity product can remain positive. The singular direction, which in earlier revisions carried **hac** as scaffolding (the finite window densities are built from **rdDeriv** against ν_i), no longer needs it; the design had flagged the hypothesis as removable, and removing it cost one inequality lemma and a null-set subtraction, because the affinity had been total since its definition (Subsection 4.1). A degenerate-case lemma is included: if any single factor pair is mutually singular, the products are (**infinitePi_mutuallySingular_of_mutuallySingular**).

4 Proof engineering

Both routes are summarized in Figure 2. The design constraint common to both: keep every estimate in $\mathbb{R}_{\geq 0}^{\infty}$ as long as possible, because each crossing between $\mathbb{R}_{\geq 0}^{\infty}$ and $\mathbb{R}$ costs finiteness and nonnegativity side conditions, and those crossings are where formalized analysis stalls.

4.1 The singular direction: cylinder squeeze and Borel–Cantelli

Suppose the infimum of partial affinity products is 0. For each k extract a finite window s_k with $\prod_{i \in s_k} H_i < 4^{-k}$, and consider the cylinder event $B_k = \{x \mid Z_{s_k}(x) < 4^{-k}\}$ where $Z_s(x) = \prod_{i \in s} \frac{d\mu_i}{d\nu_i}(x_i)$; the Radon–Nikodym derivative is total in Mathlib, so Z_s needs no hypothesis to exist. Let T be the union, over the windows s_k , of the coordinate cylinders over carriers of the singular parts $(\mu_i).\text{singularPart}(\nu_i)$; only countably many coordinates ever appear, so T is ν^{∞} -null. Two one-line estimates squeeze the shaved events $B_k \setminus T$ from both sides:



■ **Figure 2** The two proof routes. Here $H_i = \text{hellingerAffinity}(\mu_i)(\nu_i)$, Z_s is the finite-window likelihood ratio, W the ratio over a window disjoint from t , and μ^∞, ν^∞ the infinite product measures.

244 ■ $\mu^\infty(B_k \setminus T) \leq 4^{-k} \leq 2^{-k}$, because off T each μ_i restricts to its absolutely continuous
 245 part, so the μ^∞ -mass of a cylinder is at most the ν^∞ -integral of Z_{s_k} over it (**infinitePi_**
 246 **cylinder_diff_le_setLIntegral_rnDeriv**, file **Cylinder.lean**), and the integrand is
 247 below 4^{-k} there;

248 ■ $\nu^\infty(B_k^c) \leq 2^{-k}$, by Markov's inequality applied to $\sqrt{Z_{s_k}}$ together with the moment
 249 identity $\int \sqrt{Z_s} d\nu^\infty = \prod_{i \in s} H_i$ (**lintegral_prod_rnDeriv_rpow_infinitePi**), which
 250 is hypothesis-free because the affinity identity of Subsection 3.2 is.

251 Both bounds are geometrically summable, so the Borel-Cantelli lemma applies twice: $S =$
 252 $\limsup_k (B_k \setminus T)$ is μ^∞ -null, and its complement, contained in $\limsup_k (B_k \setminus T)^c$, is ν^∞ -null.
 253 The set S witnesses mutual singularity directly. Under $\mu_i \ll \nu_i$ the set T is empty and the
 254 argument reduces to the plain squeeze on B_k .

255 This route uses no martingale convergence and no zero-one law, although both exist in
 256 Mathlib and were mapped as fallbacks during design. The payoff is concrete: the entire
 257 proof is elementary integration against cylinders, and it is the reason the statement holds
 258 for arbitrary ι , since only a *sequence* of finite windows is ever extracted from the vanishing
 259 infimum.

260 One methodological remark. The first implementation carried $\mu_i \ll \nu_i$, because the
 261 cylinder density *identity* genuinely needs it: singular parts push μ^∞ -mass above the density
 262 integral, and the checker refuses the equality without them. What the squeeze consumes,
 263 however, is only the $\leq$ direction, and the excess mass lands inside a ν^∞ -null set. Asking
 264 which direction of a refused identity the proof actually uses, and where the failure of the
 265 other direction lives, turned a scoping compromise into the stronger theorem; the total
 266 definition of the affinity, fixed three milestones earlier, had already paid for the moment
 267 identity, so the generalization cost one inequality lemma and a null-set subtraction.

4.2 The absolutely continuous direction: quantitative comparison

The classical route builds $\lim_n \sqrt{Z_n}$ in $L^2(\nu^\infty)$ and identifies its square as the density of μ^∞ . Formalizing it means moving between $\mathbb{R}_{\geq 0}^\infty$ -valued densities and real-valued L^2 functions repeatedly. The implemented route instead compares measures directly. Its core is a quantitative estimate (`Equivalence.lean`):

```

theorem infinitePi_cylinder_le_withDensity_add [DecidableEq ι]
  (hac : ∀ i, μ i <= ν i) {t u : Finset ι} (htu : t ⊆ u)
  {S : Set (Π i : u, X i)} (hS : MeasurableSet S) :
  Measure.infinitePi μ (cylinder u S)
    ≤ (Measure.infinitePi ν).withDensity
      (fun x => ∏ i ∈ t, (μ i).rnDeriv (ν i) (x i)) (cylinder u S)
      + 2 * (2 - 2 * ∏ i ∈ u \ t, hellingerAffinity (μ i) (ν i)) ^ (1 / 2 : ℝ)

```

On any cylinder, μ^∞ exceeds the finite-window density measure $\nu^\infty.\text{withDensity } Z_t$ by at most an error controlled by the affinity product over the coordinates beyond t . The proof is the classical second-moment computation in disguise: with W the likelihood ratio over $u \setminus t$ and $c = \sqrt{W}$, the factorization of window moments over disjoint windows (`integral_prod_mul_prod_infinitePi`) gives $\int Z_t (c-1)^2 d\nu^\infty \leq 2 - 2 \prod_{i \in u \setminus t} H_i$, and two applications of Cauchy-Schwarz for the lower Lebesgue integral convert this into the additive bound. Every step is in $\mathbb{R}_{\geq 0}^\infty$; subtraction is truncated, and three small helper lemmas manage the truncation arithmetic.

The extension from cylinders to all measurable sets is a symmetric-difference approximation: measurable cylinders form a set algebra generating the product σ -algebra, and Mathlib's `MeasureDense` machinery shows every measurable set is approximated in measure by cylinders. A short lemma (`measure_le_add_of_le_add_on_setAlgebra`) transports the bound “ $m_1 \leq m_2 + \beta$ on a generating algebra” to all measurable sets. Given $\nu^\infty(E) = 0$ and $\varepsilon > 0$, summability of the deficits supplies, through `Summable.vanishing`, a finite window t whose tail contributes less than ε ; the density measure vanishes on E because ν^∞ does, and $\mu^\infty(E) \leq \varepsilon$ follows. The only crossing from $\mathbb{R}_{\geq 0}^\infty$ to $\mathbb{R}$ in the whole direction is the Weierstrass bridge inside the tail estimate.

Countability of ι is again never used: a summable deficit family is supported on countably many indices without any need to say so, because the argument only ever selects finite windows. The design documents anticipated a reduction of countable ι to $\mathbb{N}$ by re-indexing; the implemented route made it unnecessary, and the stated theorems are stronger than the designed ones.

4.3 Method: scout first, then implement

The development was written design-first. Before any Lean file was created, a design document fixed the definitions, the statement shapes, the milestone ladder of Figure 1, and, for every milestone, the exact Mathlib lemmas to be consumed, each verified by `grep` against the pinned Mathlib revision and recorded with file and line. The document also carried a risk register with ordered fallbacks. Implementation then proceeded one milestone at a time, each landing as a complete file with no `sorry`; the waves compiled green on the first or near-first attempt.

The register is worth reporting because two of its entries fired. The risk item for the absolutely continuous direction predicted that $\mathbb{R}_{\geq 0}^\infty/\mathbb{R}$ bookkeeping would dominate an L^2 route and prescribed keeping all cylinder identities in $\mathbb{R}_{\geq 0}^\infty$ with a single crossing; following that prescription to its end produced the comparison route of Subsection 4.2, and the L^2 machinery was never needed. The risk item for the application predicted friction in a

317 quotient-circle (`AddCircle`) model and named an interval model as fallback; the fallback
 318 was taken directly (Section 5). The residue of the one pinned-toolchain workaround that
 319 remains (a `set_option` guarding one Fubini proof) is flagged in the upstreaming plan.

320 5 Applications: the critical exponent as a phase boundary

321 The repository [12] studies the kernel $K(m, n) = \gcd(m, n) / \sqrt{mn}$ and its spectral represent-
 322 ations. One of them realizes K through Fourier coefficients of a product Poisson measure
 323 on an infinite-dimensional torus indexed by the primes, with per-prime radii $a_p = p^{-\sigma}$. The
 324 local model at each prime, kept on $\mathbb{R}$ rather than on a quotient circle, is:

```
325 noncomputable def haarLoc : Measure ℝ :=
326   (ENNReal.ofReal (2 * Real.pi))-1 • volume.restrict (Set.Ioc (-Real.pi) Real.pi)
327
328 noncomputable def poissonMeasure (a : ℝ) : Measure ℝ :=
329   haarLoc.withDensity fun θ => ENNReal.ofReal (poissonKernel a θ)
330
331
```

332 where `poissonKernel a θ` = $(1 - a^2) / |1 - ae^{iθ}|^2$ is the classical Poisson kernel, a strictly
 333 positive probability density for $0 < a < 1$.

334 5.1 The dichotomy at the critical exponent

335 A computation (`hellingerAffinity_poissonMeasure`) identifies the abstract affinity of the
 336 pair with the repository's analytic benchmark value $\text{hellinger}(a) = (2\pi)^{-1} \int_{-\pi}^{\pi} \sqrt{P_a(\theta)} d\theta$,
 337 and an analytic bridge already in the repository (`summable_hellinger_deficit_iff`) shows
 338 the deficits $1 - \text{hellinger}(p^{-\sigma})$ are summable over the primes iff $\sigma > 1/2$, with divergence at
 339 the endpoint. The mechanism is the classical one: the deficit at small a is comparable to
 340 a^2 , so the series behaves like $\sum_p p^{-2\sigma}$, which converges exactly for $\sigma > 1/2$ and diverges at
 341 $\sigma = 1/2$ like $\sum_p 1/p$.

342 Feeding this into the two directions of the dichotomy yields the phase boundary as a
 343 single statement (`PoissonDichotomy.lean`):

```
344 theorem poissonProduct_dichotomy {σ : ℝ} (hσ₀ : 0 < σ) :
345   (Measure.infinitePi (fun p : Nat.Primes => poissonMeasure ((p : ℝ) ^ (-σ)))
346     ⊥m Measure.infinitePi (fun _ : Nat.Primes => haarLoc) ↔ σ ≤ 1 / 2) ∧
347   (Measure.infinitePi (fun p : Nat.Primes => poissonMeasure ((p : ℝ) ^ (-σ)))
348     ≪ Measure.infinitePi (fun _ : Nat.Primes => haarLoc) ↔ 1 / 2 < σ)
349
350
```

351 For $\sigma > 1/2$ the two products are in fact equivalent (`poissonProduct_equivalent`); the
 352 backward implications are exclusivity, since a nonzero measure cannot be both absolutely
 353 continuous and mutually singular with respect to the same measure. The type change lands
 354 exactly at the critical exponent, with the endpoint $\sigma = 1/2$ on the singular side.

355 5.2 The translate flow, expelled without the prime number theorem

356 There is a sharper statement in the same family. For $u \in \mathbb{R}$ the log-scale flow rotates the
 357 p -th circle by $u \log p$; classically, deciding whether the translated product Poisson measure
 358 stays in the measure class of the original leads, at the endpoint $\sigma = 1/2$, to the series
 359 $\sum_p \cos(u \log p) / p$, whose boundedness is equivalent to the nonvanishing of $\zeta(1 + iu)$ — prime
 360 number theorem strength. The formalized theorem (`TranslateSingular.lean`) shows that
 361 for an explicit range of shifts no such input is needed:

```

362 theorem translate_poissonProduct_mutuallySingular {σ u : ℝ} (hσ₀ : 0 < σ)
363   (hσ : σ ≤ 1 / 2) (hu₀ : u ≠ 0)
364   (hu : |u| ≤ Real.pi / (3 * Real.log 4)) :
365   Measure.infinitePi (fun p : Nat.Primes =>
366     rotatedPoissonMeasure ((p : ℝ) ^ (-σ)) (u * Real.log p))
367   ⊥ₘ Measure.infinitePi (fun p : Nat.Primes =>
368     poissonMeasure ((p : ℝ) ^ (-σ)))
369

```

Here `rotatedPoissonMeasure` *a t* reweights the Haar slice by the shifted kernel $\theta \mapsto P_a(\theta+t)$; by 2π -periodicity this is the wrapped density of the rotated measure, so no wrap map is needed. The theorem is unconditional, includes the endpoint, and every constant in the proof is explicit and verified in-proof. The proof has three moves (`BlockPhase.lean`):

- *Deficit bound.* The abstract affinity of the rotated pair identifies with the two-point integral $H(a, t) = (2\pi)^{-1} \int \sqrt{P_a(\theta+t)P_a(\theta)} d\theta$, and the exact defect identity $1 - H(a, t) = (4\pi)^{-1} \int (\sqrt{P_a(\theta+t)} - \sqrt{P_a(\theta)})^2 d\theta$ yields, by an elementary pointwise chain, $1 - H(a, t) \geq \frac{8}{2187} a^2 (1 - \cos t)$ for $0 < a \leq 1/2$.
- *4-adic Chebyshev blocks.* Mathlib's bounds on Chebyshev's ϑ give $\vartheta(4^{k+1}) - \vartheta(4^k) \geq 4^k \log 2$ for $k \geq 7$ (dyadic blocks fail: the leading terms of the upper and lower bounds cancel exactly at ratio 2), hence $\sum_{4^k < p \leq 4^{k+1}} p^{-2\sigma} \geq \frac{1}{8(k+1)}$ for $\sigma \leq 1/2$.
- *Pigeonhole phase crossing.* The block phase $u \log p$ drifts by $\delta = u \log 4 \leq \pi/3$ per block, so every window of $\lceil 2\pi/\delta \rceil + 1$ consecutive blocks contains one whose whole phase interval lies in the arc where $1 - \cos \geq 3/2$. The good blocks carry harmonic weight, and the resulting series diverges, so $\sum_p p^{-2\sigma} (1 - \cos(u \log p))$ is not summable; the hypothesis-free singular direction (`infinitePi_mutuallySingular_of_not_summable`) concludes.

The shift range is the honest frontier of blockwise counting: beyond it one must count primes by phase within blocks, that is, use equidistribution of $u \log p$ modulo 2π , which is again zero-free-line strength and absent from the pinned Mathlib. For $|u| \leq \pi/(3 \log 4)$ the endpoint obstruction dissolves: the gate is the size of the shift, not the exponent.

Within the wider program these two theorems are the machine-checked forms of claims the accompanying essay makes about the log-scale boundary behavior of the divisibility geometry; the program's other results and its open problems are catalogued in the repository's status ledger [12] and are out of scope here.

6 Related work

Measure and probability theory have mature formalized foundations in several systems. In Isabelle/HOL, measure theory and the core of probability were developed by Hölzl and Heller [9], and infinite products of probability measures, via projective limits in the Ionescu–Tulcea style, are part of Hölzl's subsequent development [8]; the infrastructure for stating Kakutani's theorem has therefore existed in Isabelle for some time. In Coq, MathComp-Analysis provides measure construction and the Lebesgue integral [1]. In Lean, Mathlib [13] contains the Radon–Nikodym theorem, the Lebesgue decomposition, martingale convergence, and, recently, the infinite product measures this work builds on. We searched the Isabelle Archive of Formal Proofs, the Coq ecosystem, and the Lean/Mathlib libraries for a formalization of Kakutani's theorem and found none; the claim of first formalization is made to the authors' knowledge, and Section 7 commits to re-verifying it immediately before publication.

On the divergence side, Mathlib contains the Kullback–Leibler divergence, $\mathbb{R}_{\geq 0}^\infty$ -valued and standalone. A broader f -divergence framework by Degenne and Luccioli [4] exists as an external Lean development and has begun to be upstreamed; the Hellinger affinity

defined here is the f -integral at $f = \sqrt{\cdot}$, and a bridging lemma is planned for whenever the general framework lands (Section 7). The formalization of the Feldman–Hájek dichotomy for Gaussian measures [5, 6], for which the present theorem is a natural stepping stone, remains open in all systems as far as we know.

7 Artifact and upstreaming

Artifact.

The development is part of the public repository `riemann-venue` [12], Apache-2.0, frozen for this article at commit [COMMIT]. The Kakutani layer is:

File (RiemannVenue/Kakutani/)	Contents	Lines
<code>Affinity.lean</code>	Hellinger affinity and API (M1)	280
<code>FiniteProduct.lean</code>	<code>pi_withDensity</code> , integral Fubini, tensorization (M2)	143
<code>Cylinder.lean</code>	bridges, cylinder density and moment identities (M3)	364
<code>Singular.lean</code>	singular direction, hypothesis-free (M4)	258
<code>Equivalence.lean</code>	absolutely continuous direction (M5)	484
<code>Dichotomy.lean</code>	packaged dichotomy and biconditional	86
<code>PoissonSingular.lean</code>	local Poisson model, singular half of M6	217
<code>PoissonDichotomy.lean</code>	equivalence half and the packaged boundary (M6)	175
<code>SpectralMeasure.lean</code>	Bochner window factorization; spectral application	248
<code>QuadraticForm.lean</code>	Parseval face of the spectral application	311
<code>BlockPhase.lean</code>	rotated-affinity bench, 4-adic blocks, phase pigeonhole	1059
<code>TranslateSingular.lean</code>	the translate-flow singularity	297

The toolchain is Lean 4 (`leanprover/lean4:v4.32.0-rc1`) with Mathlib pinned at revision `e2361c1`. Continuous integration builds the full repository and enforces a zero-sorry policy by a textual guard in addition to the build. Axiom audits via `#print axioms` report exactly `propext`, `Classical.choice`, and `Quot.sound` on the main declarations: `infinitePi_mutuallySingular`, `infinitePi_absolutelyContinuous_of_summable`, `infinitePi_absolutelyContinuous_or_mutuallySingular`, `poissonProduct_dichotomy`, and `translate_poissonProduct_mutuallySingular`.

To build:

```
lake exe cache get && lake build
```

Upstreaming.

The general-purpose layer (everything except the Poisson application) is scheduled for Mathlib as six pull requests, about 31 public declarations in total: (1) the Hellinger affinity file; (2) `pi_withDensity` and the integral product Fubini; (3) the infinite-product/summability bridges; (4) the singular direction, in its hypothesis-free form; (5) the absolutely continuous direction together with the packaged dichotomy and the biconditional of `Dichotomy.lean`; (6) the Bochner window factorization over `infinitePi`, which is independent of the rest. PRs 1–3 and 6 have no interdependencies; 4 needs 1–3; 5 needs 4. The plan, including naming deltas, the module-system port, and known drift risks between the pin and current Mathlib, is a document in the repository (`notes/mathlib-pr-plan.md`).

441 **Verification of the novelty claim.**

442 The claim that no prior formalization exists will be re-checked against the Isabelle AFP,
 443 the Coq opam ecosystem, and the Lean/Mathlib and Zulip archives immediately before this
 444 article is published in any form, and the hedge “to the authors’ knowledge” will be retained
 445 regardless of the outcome.

446 **8 Conclusion**

447 Kakutani’s dichotomy is now a machine-checked theorem, stronger than the classical one in
 448 both index generality and hypotheses, with proof routes chosen so that the formalization
 449 consists of elementary integration against cylinders rather than martingale limit theory. The
 450 exercise supports two working hypotheses about formalizing graduate-level measure theory:
 451 the cost concentrates at value-type crossings, and route selection that minimizes those cross-
 452 ings can matter more than the availability of heavy machinery; and total definitions are
 453 cheap insurance, since the hypothesis-free singular direction and the one-sided biconditional
 454 both trace back to a definition fixed before either was in scope. Next steps are the Mathlib
 455 pull-request sequence and, once a general f -divergence framework is in Mathlib, the bridging
 456 lemma tying the affinity to it. The Feldman–Hájek dichotomy is the natural continuation.

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